

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #08 – Digital Campus Library Reservation Gateway

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1. Problem Context

The Digital Campus Library Reservation Gateway is a university library system that lets members search the physical ISBN catalog, manage dynamic holding queues for borrowed copies, automate hold notifications, and track overdue fine transactions. The two target stakeholders are the Student Member, who searches, reserves, and pays fines, and the Head Librarian, who manages the catalog, oversees overdue accounts, and can grant fine waivers.

2. Requirements Table

Five Functional Requirements (FR-001 – FR-005) and two Non-Functional Requirements (NFR-001 – NFR-002), each captured with all six required fields: ID, Type, Priority, Description, Acceptance Criteria, and Rationale.

ID	Type	Priority	Description	Acceptance Criteria	Rationale
FR-001	Functional	High	The system shall allow a Student Member to search the ISBN catalog and place a hold request on a currently loaned book, adding them to a FIFO waitlist for that title.	Pass: Member is assigned the correct sequential position in the queue. Fail: A member already holding the maximum allowed reservations (5) is permitted to place another.	Provides fair, ordered access to high-demand titles without manual librarian mediation.
FR-002	Functional	High	The system shall automatically notify a Student Member (email/SMS) within 15 minutes of a held book becoming available, and reserve the copy for that member for 48 hours.	Pass: Notification is sent within the SLA and the copy is blocked from other holds during the 48-hour window. Fail: Notification is not sent, or the copy is issued to another member during the hold window.	Prevents missed pickups and keeps the FIFO queue moving efficiently.
FR-003	Functional	Medium	The system shall calculate overdue fines on a per-day, per-book basis, cap the total fine per book, and automatically flag a member's account as restricted once total dues cross a configurable threshold.	Pass: Fine accrues correctly from the day after the due date and the account is restricted exactly when the threshold is crossed. Fail: Fine is miscalculated or restriction is not enforced.	Ensures financial accountability and incentivizes timely returns.
FR-004	Functional	Medium	The system shall allow the Head Librarian to add, edit, and retire catalog entries (title, author, ISBN, copies, shelf location).	Pass: Catalog changes are persisted and immediately reflected in search results. Fail: A duplicate or malformed ISBN is accepted into the catalog.	Keeps the catalog accurate and current for all search and hold operations.
FR-005	Functional	Low	The system shall allow a Student Member to pay overdue fines through an integrated payment gateway, and allow the Head Librarian to waive or reduce a fine with a logged reason.	Pass: A successful payment lifts the account restriction and is reflected in the ledger; a waiver is logged with librarian ID, reason, and timestamp. Fail: A payment is not reflected, or a waiver is applied without an audit entry.	Gives flexibility for legitimate exceptions while preserving a complete audit trail.
NFR-001	Performance	High	The catalog search query shall return paginated results (25 records/page) across a catalog of up to 50,000 titles	Pass: Benchmark tests confirm p95 latency under 150 ms with 500 simulated concurrent users. Fail: p95	Search responsiveness is central to a

ID	Type	Priority	Description	Acceptance Criteria	Rationale
			in under 150 ms at the 95th percentile under normal load.	latency exceeds 150 ms under the same test conditions.	usable self-service catalog experience.
NFR-002	Security & Availability	High	The system shall enforce role-based access control between Student Member and Head Librarian roles, encrypt all fine-payment traffic using TLS 1.3, and maintain 99.5% uptime during the academic semester.	Pass: Unauthorized role-elevation attempts are blocked and logged; a penetration test confirms payment traffic is encrypted; uptime meets target. Fail: Privilege escalation succeeds, or uptime falls below 99.5%.	Protects sensitive personal and financial data and ensures the library service remains available during peak exam/registration periods.

3. UML Use-Case Diagram

Primary actors (Student Member, Head Librarian) initiate use cases directly; secondary/system actors (Notification Service, Payment Gateway) support the system from outside. Place Hold Request «includes» Verify Membership Eligibility on every execution, and Apply Fine Waiver «extends» Pay Overdue Fine only under the condition that the Head Librarian approves a waiver.

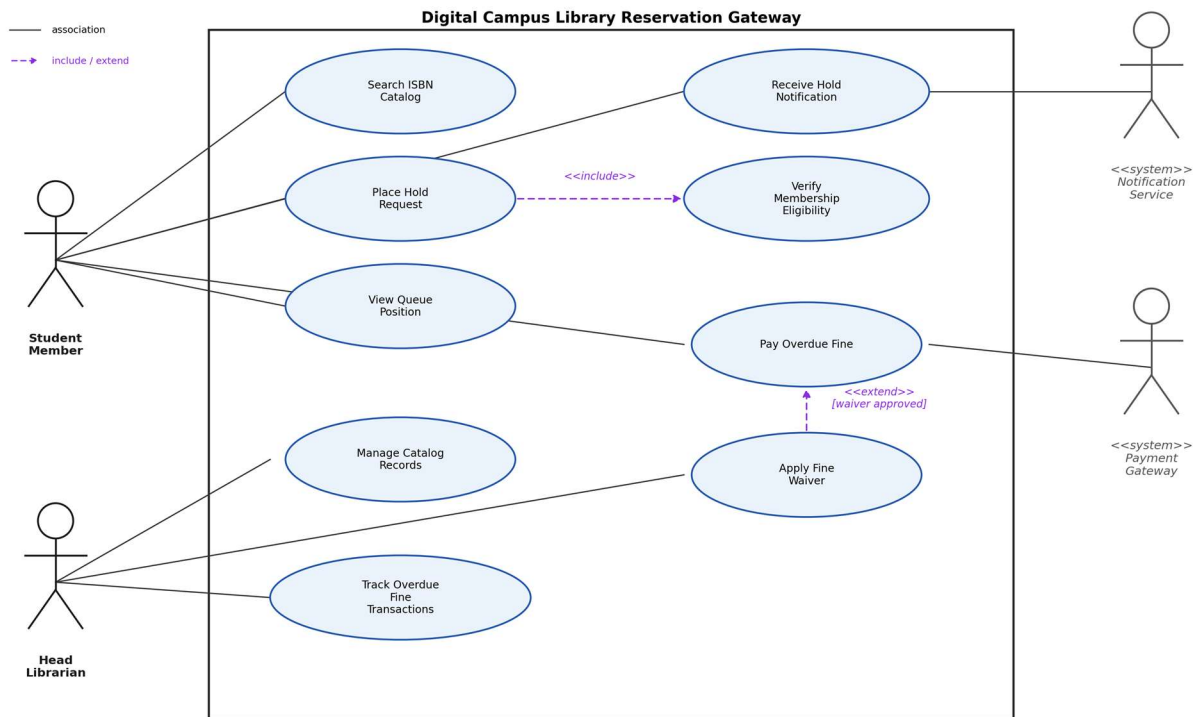


Figure 1: Use-case diagram for the Digital Campus Library Reservation Gateway

Actor Summary

- Student Member (primary) – Search ISBN Catalog, Place Hold Request, View Queue Position, Receive Hold Notification, Pay Overdue Fine.
- Head Librarian (primary) – Manage Catalog Records, Track Overdue Fine Transactions, Apply Fine Waiver.
- Notification Service (secondary/system) – delivers the automated Receive Hold Notification alert.

- Payment Gateway (secondary/system) – processes the Pay Overdue Fine transaction.

4. Use-Case Flow Specification – Place Hold Request

Use Case: Place Hold Request

Primary Actor: Student Member

Secondary Actors: None (Verify Membership Eligibility is an internal included use case)

Preconditions

- The Student Member is authenticated and logged into the system.
- The requested title exists in the ISBN catalog and is currently checked out (unavailable for immediate loan).
- The member does not already hold an active reservation on the same title.

Postconditions

- Success: The member is added to the FIFO waitlist queue at the correct position, and the hold event is logged.
- Failure: No queue entry is created and system state remains unchanged.

Main Success Scenario

1. Student Member searches for a book by title, author, or ISBN.
2. System displays matching catalog entries with real-time availability status.
3. Student Member selects a currently-loaned title and chooses “Place Hold.”
4. System performs the included use case Verify Membership Eligibility, checking active membership status, outstanding fine balance, and current hold count.
5. System confirms the member is eligible to place a hold.
6. System adds the member to the FIFO waitlist queue for that ISBN.
7. System displays a confirmation screen showing the member's queue position and estimated wait.
8. System logs the hold-request timestamp and queue position for audit and notification purposes.

Alternate Flow A1 – Member Not Eligible

Trigger: Verify Membership Eligibility fails during step 4 of the main scenario.

- 4a.** At step 4, Verify Membership Eligibility determines the member has outstanding fines above the configured threshold, or already holds the maximum of 5 active reservations.
- 4b.** System aborts the hold-placement operation before any queue entry is created.
- 4c.** System displays an error message stating the specific ineligibility reason (e.g., “Outstanding fine exceeds limit” or “Maximum holds reached”).
- 4d.** Use case ends without the member being added to the queue; the member is redirected to the fines/holds summary screen to resolve the blocking condition.